

Week 3: Wiggins

1. Some claims about sortals

(2) (p. 56): A predicate/concept is a sortal just in case 'to say that x falls under f or that x is an f is to say what x is'

D(i) Everything (every substance) falls under some sortal at every time at which it exists

D(ii) For every thing, there is some sortal under which it falls at every time at which it exists.

(*Substance-sortals*: sortals which apply to a thing at every moment of its existence, if they apply to it at any moment at all. *Phased-sortals*: sortals which are not substance sortals.)

Many ordinary count nouns are sortals.

2. Coincidence under a sortal

According to Wiggins, whenever f is a sortal, there is a relation ' x is the same f as y ', or ' x coincides with y under f in the manner of coincidence required for members of f ', which [necessarily] holds between x and y just in case $x=y$ and x is an f (and y is an f).

Obviously this claim is trivial unless we require this relation to obey some further conditions.

What could these conditions be? When we look at Wiggins' examples, we see that he's imagining the relation in the case of 'person' as involving a certain distinctive sort of spatiotemporal continuity. The thought seems to be that this sort of spatiotemporal continuity holds between *people* only when they're identical, although it may hold between distinct objects of other sorts—for example, between me and the aggregate of the atoms that composed me in 1973. For x to be the same person as y is for x and y to be people, and for x and y to stand in this specific relation of spatiotemporal continuity.

I suggest that spatiotemporal continuity (of a given variety) is best made sense of as a *four-place* relation between pairs of objects and pairs of times. Pretending that time is discrete, we might say something like this: x at t is spatiotemporally continuous in such-and-such way with y at $t+n$ just in case there's a sequence of objects $a_1, \dots, a_{n-1}$ such that x at t is similar, in respect of location, shape, etc., to a_1 at $t+1$, and a_i at $t+i$ is similar in those respects to a_{i+1} at $t+i+1$, and a_{n-1} at $t+n-1$ is similar in those respects to y at $t+n$.

(We *could* try construing spatiotemporal continuity relations as two-place relations that hold absolutely, without reference to any specific times. But how would this go? Does a spatiotemporally discontinuous object count as being spatiotemporally continuous with itself? Do two objects which overlap at any time automatically count as being spatiotemporally continuous?...)

This suggests that we might understand Wiggins's claim according to the following schema:

- (S) If F is a sortal, there is some quaternary relation R , meeting certain conditions, such that the inferences

$$\frac{\begin{array}{l} Rab_1t_2 \\ Fat_1 \\ Fbt_2 \end{array}}{a = b} \qquad \frac{\begin{array}{l} Fat_1 \\ a \text{ exists at } t_2 \end{array}}{Ra_1t_2}$$

are valid.

What further conditions do we require the relation Rab_1t_2 to satisfy? Here are some relations that I doubt would be acceptable to Wiggins:

$a=b$

a is a member of all the same sets as b

a is colocated with b at every time at which a and b exist

a is colocated with b at some time

a is colocated with b at t_1

a is colocated with b at t_1 and t_2

What feature do spatiotemporal continuity relations (as imagined above) have, that these relations lack? The obvious answer: the only facts about a and b that matter to whether a at t_1 is spatiotemporally continuous with b at t_2 are the facts about what a and b are like *at t_1 and t_2* , respectively: we don't help ourselves to information about the careers of a and b at other times. The spatiotemporal continuity relations, in other words, are *temporally localised*:

A four-place relation R between pairs of objects and times is *temporally localised* just in case necessarily, if Rab_1t_2 , and a is co-located at t_1 with a' , and b is co-located at t_2 with b' , it follows that $Ra'b't_1t_2$.

(Perhaps 'co-located' should be replaced here by 'mereologically coincident'?)

If we take the restriction in (S) to be a restriction to temporally localised relations, the resulting principle is equivalent to the following:

- (Exclusivity) If F is a sortal, the inference

$$\frac{\begin{array}{l} \text{Fat} \\ \text{Fbt} \\ \text{a is colocated at t with b} \end{array}}{a = b}$$

is valid.

Proof: to derive Exclusivity from (S), suppose Fat, Fbt, and a is colocated at t with b. By the right-hand inference, Raatt. By the temporal localisation of R, Rabtt. So by the left-hand inference, a=b. To derive (S) from Exclusivity, let Rabt₁t₂ be the relation $\Box x(\text{Fxt}_1 \ \& \ \text{Fxt}_2 \ \& \ x \text{ is colocated at } t_1 \text{ with a} \ \& \ x \text{ is colocated at } t_2 \text{ with b})$.

3. The 'only a and b' rule.

'In notionally pursuing object a in order to ascertain its coincidence or non-coincidence with b, or in retracing the past history of b to ascertain its identity link with a, I ought not to need to concern myself with things that are other than a or other than b. This is not to deny that singling a out or singling b out may involve contrasting a or b with a background of other things. But the identity of a with a, or b with b, and of a with b, once we are clear which things a and b are, ought to be a matter strictly between a and b themselves'.

This principle is used to argue against the view that a ship survives replacement of planks only if the discarded planks are not retained and rearranged shipwise, as in Hobbes's story.

The last sentence of the quoted claim makes it look like Wiggins thinks that this principle follows from the claim that identity is intrinsic, or something like that. But consider: in determining whether the inventor of bifocals was the first Postmaster-General, I certainly do need to concern myself with things that are other than the inventor of bifocals and other than the first Postmaster-General. Why shouldn't it be similar with descriptions like 'the ship that was in the harbour at t₁'?

Interpretative Suggestion: Suppose that a spatiotemporal path P is traced out by some f (f a sortal). And suppose that Q is a spatiotemporal path that (i) is an intrinsic duplicate of P, and (ii) has at least one time-slice in common with a path traced out by an f. Then Q is traced out by an f.

Wiggins's suggested solution to the Ship of Theseus puzzle: an artifact can't survive the loss of more than half of its original matter.

Problem: what about the new artifacts that come into existence when old artifacts are subjected to too much replacement of parts? Don't they violate the 'only a and b' rule?

4. What's really going on?

A natural picture: some predicates are satisfied by an object at a time in virtue of facts about what that object is like at other times. EG: 'such that it is red at 12.00 pm on Sept. 12, 2003'; 'such that it is going to be square three hours hence'. Typical sortal predicates are like this. For something to be a statue, say, is (in part at least) for its whole spatiotemporal career to hang together in a certain way, and to contrast in a certain way with its spatiotemporal surroundings. For example, if there's going to be an arm-replacement in x 's future, then x is a statue partly in virtue of the fact that x will survive this arm-replacement.

Suggestion: maybe Wiggins wants to reject this picture about the order of determination. It's not that x is a statue in virtue of being something that will survive the arm-replacement; rather, x is something which will survive the arm-replacement in virtue of being a statue. In general, things have properties like 'being going to survive an arm-replacement' or 'having been in school in 1953' partly in virtue of their sortal properties.

This can be seen as giving some motivation to Exclusivity. If things get to have the careers they have in virtue of their sortal properties, it would be natural to conclude that things that are alike in their sortal properties and in their other present properties, then they must also be alike in their spatiotemporal careers. So if we think that there can't be two Fs that are colocated throughout their careers, we'll be pushed to accept Exclusivity.

I can see how this view might make sense to a *presentist*: someone who thinks that only the present is real. (And maybe even to other "A-theorists", who think that the present is metaphysically special.) But from an *eternalist*, *B-theoretic* point of view—from the point of view who thinks that the present is just one among many other times, in no way metaphysically special—it seems problematic. If the present isn't metaphysically special, then properties like 'being square now' are no different from properties like 'being square three hours hence': if things get to have the latter property only in virtue of falling under a certain sortal, then they must also get to have the former property only in virtue of falling under a certain sortal. But how could this work? What prospect is there of a *general* analysis of ' x is square at t ' which is of the form 'There is some sortal f such that x is an f and ... f ... x ... t ...'?